Perfect-Power Obstructions and Sparse Powerful Values in Cyclic Decimal Concatenations

Qichao Wang
Hebei University of Technology
Daoyu Dong
University of Electronic Science and Technology of China

October 27, 2026

Abstract

We prove a quantitative density-zero bound for powerful values among cyclic concatenations of initial positive integers, uniformly in the rotation and in every fixed integer base. At the decimal boundaries $m = 10^d - 1$, exact valuations and finite-field characters exclude every rotation for a set of digit lengths of lower density exceeding 95.53%. All boundaries $1 \le d \le 27$, as well as $d = 54, 243$, are excluded. Explicit period formulas give further infinite families. A general prime-adic lifting theorem yields exact valuations and local fifth- and seventh-power equivalences for every cut, including positive valuations at the new primes.

1 Statements

In base $g \ge 2$, let W_m concatenate $1, 2, \ldots, m$, and let $D(m)$ be its digit length, with $W_0 = D(0) = 0$. For $0 \le k < m$, moving the first k whole integer blocks to the end gives

$$R_{m,k} = g^{D(k)} W_m - (g^{D(m)} - 1) W_k. \tag{1}$$

In base ten these numbers, in increasing order, form OEIS A001292 [1]. Kashihara conjectured that none except 1 is a perfect power. Ripà [2] excludes $m \equiv 2, 3, 5, 6 \pmod 9$ by a digit-sum argument. The full conjecture remains open. We give a uniform sparsity bound and local obstructions for complete digit boundaries.

Recall that a positive integer is *powerful* if every prime divisor has exponent at least two. Every nontrivial perfect power is powerful. Logarithms in the following theorem and in Section 5 are natural.

Theorem 1 (Quantitative sparsity)**.** *For every fixed integer base $g \ge 2$,*

$$\#\{(m,k) : 2 \le m \le X,\ 0 \le k < m,\ R_{m,k} \text{ powerful}\} \ll_g \frac{X^2 (\log\log X)^2}{\log X}. \tag{2}$$

For A001292, the corresponding count among the first N terms is $O(N(\log\log N)^2/\log N)$.

For the next result the base is ten and v_3 denotes the 3-adic valuation.

Theorem 2 (Complete digit boundaries)**.** *Let $d \ge 1$, $m = 10^d - 1$, $h = v_3(d)$, and $u = d/3^h$. Every rotation satisfies*

$$v_3(R_{m,k}) = h + 2, \qquad \frac{R_{m,k}}{3^{h+2}} \equiv u(2 + 6d) \pmod 9. \tag{3}$$

Corollary 1. *Put $v = h + 2$ and let $\mathcal{P}(v)$ be its set of prime divisors. No rotation at $m = 10^d - 1$ is a perfect power in any of the following cases:*

$\mathcal{P}(v)$	*condition on u*
$\{2\}$	$u \equiv 1 \pmod 3$
$\{3\}$	$u \equiv 1, 2, 7, 8 \pmod 9$
$\{2, 3\}$	$u \equiv 1, 7 \pmod 9$.

This includes $d \equiv 3, 6, 9, 21, 24 \pmod{27}$ and $d = 81$.

Proof. A prime exponent of a perfect power must divide v. A square requires v even and its normalized unit congruent to 1 modulo 3; by (3) this requires $u \equiv 2 \pmod 3$. A cube requires $3 \mid v$, hence $h \geq 1$ and $3 \mid d$. Its normalized unit must be $\pm 1 \pmod 9$, so $2u \equiv \pm 1 \pmod 9$, or $u \equiv 4, 5 \pmod 9$. The table excludes every available prime exponent. □

For the additional boundary primes, put

$$(p, s) \in \{(37, 3), (333667, 9)\}, \qquad C_s = \frac{9(11 - 10s)}{200} \pmod p.$$

Define

$$A_0 = 0, \qquad A_r = -9 \sum_{j=1}^{r} \frac{10^{j-1}}{10^j - 1} \pmod p \quad (0 < r < s).$$

All denominators here are units modulo p; $C_3 = 33$ and $C_9 = 221885$.

Theorem 3 (Two affine residue formulas)**.** *For every $d \geq 1$, write $d = sn + r$, $0 \leq r < s$. Then*

$$V_d := W_{10^d-1} \equiv C_s n + A_r \pmod p, \qquad R_{10^d-1,k} \equiv 10^{D(k)} V_d \pmod p. \tag{4}$$

Theorem 4 (General boundary-prime lifting)**.** *Let $p \geq 7$ be prime, $r = \operatorname{ord}_p(10)$, and choose P with*

$$r \mid P, \qquad 10^P \equiv 1 \pmod{9r}, \qquad p \nmid P, \qquad C := V_P \not\equiv 0 \pmod p.$$

For every positive d divisible by P, set $a = v_p(d)$. Every whole-block rotation satisfies

$$v_p(R_{10^d-1,k}) = a, \qquad \frac{R_{10^d-1,k}}{p^a} \equiv 10^{D(k)} C \frac{d}{P p^a} \pmod p. \tag{5}$$

In particular, $d = Ppt$ with $p \nmid t$ excludes all perfect powers. More generally, every possible prime exponent divides both $v_3(d) + 2$ and $v_p(d)$, for each applicable pair (p, P). The six primes in the table of Section 4 all satisfy the hypotheses. For the two affine formulas take $P = s$ and $C = C_s$. If a prime exponent ℓ divides $(p - 1)/r$, then $Cd/(Pp^a)$ must also be an ℓth power modulo p. In particular, $d = 111t$ with $37 \nmid t$ and $d = 3003003t$ with $333667 \nmid t$ are excluded. The new choice $p = 757, P = 27$ also excludes $d = 327024 = 27 \cdot 757 \cdot 16$, since the valuations of every rotation at 3 and 757 are respectively 5 and 1.

Corollary 2 (Density of excluded complete boundaries)**.** *The set of positive integers d for which every rotation at $m = 10^d - 1$ is not a nontrivial perfect power has lower natural density at least*

$$\rho := \frac{28659820221646166879732231 30}{29999999999999999999999999997} = 0.9553273407\ldots.$$

Theorem 5 (Further complete-boundary exclusions)**.** *No rotation at $m = 10^d - 1$ is a nontrivial perfect power when $1 \leq d \leq 27$, or when $d = 54$ or 243. The same conclusion holds for*

$$\begin{aligned} &d \equiv 12, 15 \pmod{333}, \\ &d \equiv 27 \pmod{37249072371}, \qquad d \equiv 243 \pmod{25475666076}. \end{aligned} \tag{6}$$

The finite inputs to this theorem are exact modular certificates; Section 4 gives their verification and the unbounded period argument.

2 Exact 3-adic values at complete boundaries

Write $V_d = W_{10^d-1}$ and $S(d) = D(10^d - 1)$. We will prove the stronger intermediate congruence

$$\frac{V_d}{9d} \in \mathbb{Z}_3, \qquad \frac{V_d}{9d} \equiv 2 + 6d \pmod 9. \tag{7}$$

Quotients by d in this section are taken in $\mathbb{Q}_3$.

Lemma 1 (Averaging power series)**.** *For integers $d \geq 1$ and $n \geq 1$,*

$$v_3\left(\frac{1}{d}\sum_{j=1}^{d} j^n\right) \geq -\lfloor \log_3(n+1) \rfloor. \tag{8}$$

Suppose $f(X) = \sum_{n\geq 0} c_n X^n$ has $v_3(c_n) \geq n$, $c_0 \equiv 5 \pmod 9$, $c_1 \equiv 3 \pmod 9$, and $v_3(c_2) \geq 3$. Then

$$\frac{1}{d}\sum_{j=1}^{d} f(j) \equiv 2 + 6d \pmod 9. \tag{9}$$

Proof. Using Stirling numbers of the second kind gives the exact identity

$$\frac{1}{d}\sum_{j=1}^{d} j^n = \sum_{r=0}^{n} \left\{ {n+1 \atop r+1} \right\} \frac{r!}{r+1}\binom{d-1}{r}.$$

Each denominator loses at most $\lfloor \log_3(n+1) \rfloor$ factors of 3, proving (8). For $n \geq 3$, $n - \lfloor \log_3(n+1) \rfloor \geq 2$, so all these terms vanish modulo 9 after averaging. The same holds for $n = 2$ by its stronger coefficient bound. Only $c_0 + c_1(d+1)/2$ remains, congruent to $5+3(d+1)/2 = 2+6d \pmod 9$. The coefficient bounds also justify convergence and termwise averaging. □

Define the 3-adic logarithm explicitly by

$$\ell = \sum_{r\geq 1} \frac{(-1)^{r+1} 9^r}{r}, \qquad v_3(\ell) = 2.$$

In the following power series put

$$E = e^{\ell X}, \quad a = E/10, \quad q = 9a, \quad b = E - 1, \quad S(X) = \frac{(9X-1)E + 1}{9},$$

and define

$$\frac{U(X)}{9} = a^2 + \frac{ab+1}{9}\sum_{r\geq 0}\binom{q}{r+2} b^r, \qquad f(X) = \frac{U(X)}{9} e^{-\ell S(X)}. \tag{10}$$

At every positive integer j, $S(j)$ is the digit length of V_j, and $U(j)$ concatenates all j-digit integers. Indeed, with $a = 10^{j-1}$, $b = 10^j - 1$, and $q = 9a$, the usual arithmetic-geometric sum gives

$$b^2 U(j) = (ab+1)(1+b)^q - ((1+b)b + 1).$$

Expanding $(1+b)^q$ cancels its constant and linear terms and gives (10), since $a + q = 10^j$ and $aq = 9a^2$.

Lemma 2. *The coefficients of f satisfy all hypotheses of Lemma 1.*

Proof. Let $\mathcal{A}$ consist of series whose degree-n coefficient has valuation at least n. It is closed under addition and multiplication. In the rth summand of $U/9$, the factor b^r makes all degrees below r zero. The factor q in $\binom{q}{r+2}$ cancels the displayed denominator 9. The remaining expression is a polynomial in E with 3-adic denominators bounded by $(r+2)!$. Its degree-n coefficient, for $r \leq n$, therefore has valuation at least

$$2n - v_3(n!) - v_3((r+2)!) \geq n.$$

Here $v_3(n!) \leq (n-1)/2$ for $n \geq 1$, and $v_3((r+2)!) \leq (n+1)/2$; the degree-zero case is immediate. The term a^2 satisfies the same bound. Thus $U/9 \in \mathcal{A}$.

The constant term of S is zero. Its linear coefficient is $1 - \ell/9$, of valuation at least 2, and for $n \geq 2$ its coefficient is $\ell^{n-1}(n - \ell/9)/n!$, of valuation at least n. Hence $S \in \mathcal{A}$. Expanding $e^{-\ell S}$ preserves these bounds: its rth exponential term gains $2r - v_3(r!) \geq 0$ factors of 3. Consequently $f \in \mathcal{A}$.

Expansion through degree two in (10) gives

$$\begin{aligned} c_0 &= \frac{1}{200}, \\ c_1 &= \frac{169\ell}{3000} + \frac{\ell^2}{1800}, \\ c_2 &= \frac{671\ell^2}{80000} + \frac{353\ell^3}{54000} + \frac{\ell^4}{32400}. \end{aligned}$$

Since $\ell/9 \equiv 1 \pmod 9$, these imply $c_0 \equiv 5 \pmod 9$, $c_1 \equiv 3 \pmod 9$, and $v_3(c_2) \geq 3$. □

Proof of Theorem 2. The concatenation of the complete digit blocks yields

$$V_d = 10^{S(d)} \sum_{j=1}^{d} U(j) 10^{-S(j)} = 9\, 10^{S(d)} \sum_{j=1}^{d} f(j).$$

Lemma 1 and $10^{S(d)} \equiv 1 \pmod 9$ prove (7). Its right side is a 3-adic unit, so $v_3(V_d) = h + 2$.

To transfer the normalized unit to every rotation, note that

$$S(d) = \frac{9d(d+1)}{2} \left(1 + \sum_{r=1}^{d-1} \frac{2 \cdot 9^r}{r+2} \binom{d-1}{r} \right). \tag{11}$$

This follows by expanding $S(d) = 9 \sum_{j=1}^{d} j 10^{j-1}$ in powers of 9. Each term after 1 in the bracket is divisible by 3; hence $v_3(S(d)) = 2 + h + v_3(d+1)$. Lifting the exponent gives

$$v_3(10^{S(d)} - 1) = 4 + h + v_3(d+1).$$

In (1), $10^{D(k)} \equiv 1 \pmod 9$ and the second term is divisible by 3^{h+4}. Thus $R_{m,k} \equiv V_d \pmod{3^{h+4}}$. Combining this with (7) proves both assertions in (3). □

3 Lifting at boundary primes

Lemma 3 (Boundary periods)**.** *Let $q > 3$ be prime, $r = \operatorname{ord}_q(10)$, and $\gcd(r, 10) = 1$. Choose P with $r \mid P$ and $10^P \equiv 1 \pmod{9r}$. Then $S(d) \pmod r$ has period P, and $V_d \pmod q$ has period qP. More precisely, for $t \geq 0$ and $0 \leq s < P$,*

$$V_{tP+s} \equiv t V_P 10^{S(s)} + V_s \pmod q. \tag{12}$$

Proof. In a complete j-digit block let $a = 10^{j-1}$, $B = 10^j$, and $N = 9a$. Appending that block acts by $z \mapsto B^N z + U(j)$. Modulo q, if $B \neq 1$,

$$U(j) = \frac{(a(B-1)+1)B^N - (B(B-1)+1)}{(B-1)^2}; \tag{13}$$

if $B = 1$, it is $Na + N(N-1)/2$. The affine map therefore depends only on $a, B, N \pmod q$ and $N \pmod r$. These data repeat after P steps: $r \mid P$ fixes the first three, and $10^P \equiv 1 \pmod r$ fixes the last. Also $9j10^{j-1} \pmod r$ has period P, and

$$S(P) = \frac{(9P-1)10^P + 1}{9} \equiv 0 \pmod r$$

by the stronger modulus $9r$. Thus $S(d) \pmod r$ is periodic, and a complete P-cycle acts by $z \mapsto z + V_P$ modulo q. Appending s more blocks proves (12), and adding q cycles proves the asserted period of V_d. □

Proof of Theorem 4. We interpolate a complete P-cycle of digit blocks and average it over an arbitrary number of cycles, retaining division by that number. Put

$$\lambda = \frac{\log(10^r)}{r} \in p\mathbb{Z}_p, \qquad \omega = 10e^{-\lambda}, \qquad L = P\lambda.$$

Then $\omega^r = 1$, and its reduction has order r. For $s = 1, \ldots, P$ and $X \in \mathbb{Z}_p$, set

$$j = PX + s, \quad B = 10^s e^{LX}, \quad A = B/10, \quad N = 9A, \quad b = B - 1, \quad S_s(X) = \frac{(9j-1)B + 1}{9}.$$

Let $E_s = 9s10^{s-1}$ and $H_s = S(s)$ be fixed integers. Define

$$U_s(X) = \frac{(Ab+1)\omega^{E_s} e^{\lambda j N} - (Bb+1)}{b^2}, \qquad F(X) = \sum_{s=1}^{P} U_s(X) \omega^{-H_s} e^{-\lambda S_s(X)}.$$

For integral $X \geq 0$, the period conditions give $jN \equiv E_s \pmod r$ and $S(j) \equiv H_s \pmod r$. Thus $U_s(X)$ is the actual j-digit block and $F(X) = \sum_{j=PX+1}^{P(X+1)} U_j 10^{-S(j)}$. For $r \mid s$, remove the apparent singularity by

$$U_s(X) = AN + (Ab+1) \sum_{t \geq 0} \binom{N}{t+2} b^t. \tag{14}$$

Here $b \in p\mathbb{Z}_p$ throughout $\mathbb{Z}_p$; if $r \nmid s$, b is a unit.

Write $F(X) = \sum_{n\geq 0} c_n X^n$. The coefficient bound is

$$v_p(c_n) \geq n - v_p(n!) - \left\lfloor \frac{n+1}{p-1} \right\rfloor \geq \left\lceil \frac{p-3}{p-1} n \right\rceil \quad (n \geq 1). \tag{15}$$

To see this, each nth derivative of e^{LX} gains at least p^n. The same holds for the nth derivatives of $\lambda j N$ and $-\lambda S_s$, for $n \geq 1$: a derivative of the linear factor j loses at most one L, compensated by λ. Exponentiating preserves the property, as do inverses of unit denominators. In the tth summand of (14), an undifferentiated factor b contributes p, while a factor differentiated $i \geq 1$ times contributes p^i. Combining all factors gives at least $p^{\max(t,n)}$ for the nth derivative, before the denominator $(t+2)!$. It loses at most $\lfloor (t+1)/(p-1) \rfloor$ powers of p. For $t \leq n$ this gives the asserted bound, and for $t > n$ use that $t - \lfloor (t+1)/(p-1) \rfloor$ is nondecreasing. The tails converge with all derivatives. The normalizing exponential preserves the bound; division by $n!$ gives the first inequality. The second uses $v_p(n!) \leq (n-1)/(p-1)$.

For $M \geq 1$, the exact Stirling identity

$$\frac{1}{M} \sum_{i=0}^{M-1} i^n = \sum_{t=0}^{n} \left\{ {n \atop t} \right\} \frac{t!}{t+1} \binom{M-1}{t}$$

bounds its valuation below by $-\lfloor \log_p(n+1) \rfloor$. For $p \geq 7$, (15) exceeds this loss by at least one: if $k = \lfloor \log_p(n+1) \rfloor \geq 1$, then $n \geq p^k - 1$ makes its final bound at least $(p-3)k \geq k+1$; the case $k = 0$ is immediate. Termwise averaging therefore gives

$$\frac{1}{M} \sum_{i=0}^{M-1} F(i) \equiv c_0 \equiv C \pmod{p}.$$

The last equality follows by concatenating the first P blocks and using $r \mid S(P)$. Likewise $V_{PM} = 10^{S(PM)} \sum_{i=0}^{M-1} F(i)$ and $r \mid S(PM)$ give the exact valuation $v_p(M)$ and normalized unit $CM/p^{v_p(M)}$.

Finally put $d = PM$ and $\nu = v_p(10^r - 1) \geq 1$. Lifting the exponent gives $v_p(10^d - 1) = \nu + a$. In $S(d) = d10^d - (10^d - 1)/9$, the summands have unequal valuations a and $\nu + a$, so $v_p(S(d)) = a$. Since $r \mid S(d)$,

$$v_p(10^{S(d)} - 1) = \nu + a \geq a + 1.$$

The rotation identity (1) now proves both assertions in (5), including the factor contributed by the cut. □

Proof of Theorem 3. For both pairs in Theorem 3,

$$\operatorname{ord}_p(10) = s \mid 9, \qquad v_p(10^s - 1) = 1, \qquad S(j) := D(10^j - 1) \in 9\mathbb{Z}.$$

Let U_j concatenate the j-digit integers, and put $B = 10^j$, $b = B - 1$, $a = B/10$, $q = 9a$. Its length jq is divisible by s. If $s \nmid j$, the block formula used in (10) gives $U_j \equiv -9a/b \pmod{p}$. If $s \mid j$, its multiplier B is one modulo p, so summing the appended integers gives $U_j \equiv 9/200 \pmod{p}$. Thus $U_j \pmod{p}$ has period s. Pairing each nonidentity sth root $z = 10^j$ with z^{-1} yields

$$\sum_{j=1}^{s-1} \frac{10^j}{10^j - 1} = \frac{s-1}{2}, \qquad \sum_{j=1}^{s} U_j \equiv -\frac{9(s-1)}{20} + \frac{9}{200} = C_s \pmod{p}.$$

All block multipliers are one modulo p, proving (4), including its rotation assertion. This proves Theorem 3. Its corresponding valuations and normalized units follow from Theorem 4 with $P = s$.

□

4 Power characters and periodic transfer

For a prime q and a prime $e \mid q - 1$, put $\chi_{q,e}(z) = z^{(q-1)/e} \pmod{q}$. A nonzero eth power has character 1. If

$$10^{S(d)} \equiv 1 \pmod{q}, \qquad \chi_{q,e}(10) = 1, \qquad \chi_{q,e}(V_d) \notin \{0, 1\}, \tag{16}$$

then (1) gives the same nontrivial character for every $R_{10^d-1,k}$. Thus none is an eth power. When $v_3(d) + 2$ is a power of e, Theorem 2 eliminates all other prime exponents.

Here are the finite inputs. Their nonzero cycle constants also verify every hypothesis of Theorem 4:

i	q_i	r_i	P_i	$V_{P_i} \pmod{q_i}$
1	37	3	3	33
2	757	27	27	538
3	333667	9	9	221885
4	440334654777631	27	27	178343635184937
5	1527791	43	903	547718
6	43037	29	812	409

For each row of the next table, $r_i \mid S(d)$, $\chi_{q_i,e}(10) = 1$, and the displayed character is nonzero and different from 1. Each $v_3(d) + 2$ is a power of the listed e.

d	e	i	$\chi_{q_i,e}(V_d)$
$2, 14, 20$	2	2	-1
$5, 26$	2	3	-1
$8, 11, 17, 18$	2	1	-1
$12, 15$	3	1	26
23	2	4	-1
27	5	5	990998
54	5	4	386179030766766
243	7	6	12819

The accompanying `BOUNDARY_CHARACTERS.json` records these exact inputs; `verify_boundary_characters.py` verifies them. Their finite arithmetic consists of complete factorizations of $q_i - 1$, Lucas primality witnesses, modular powers, and the recurrence $V_j = B^N V_{j-1} + U(j)$ with (13). An independent verification uses exact matrix powering for every listed d. For a Lucas witness a, the checked conditions are $a^{q-1} \equiv 1 \pmod{q}$ and $\gcd(a^{(q-1)/\ell} - 1, q) = 1$ for every prime $\ell \mid q - 1$; the factor primes are verified by trial division. These conditions force every prime divisor of q to be at least q, proving primality.

Proof of Theorem 5. Corollary 1 and the table exhaust $d = 1, \dots, 27$; the last two rows handle 54 and 243. Equation (16) excludes every available prime exponent.

For any row put $h = v_3(d)$ and $T = q_i \operatorname{lcm}(P_i, 3^{h+1})$. Replacing d by $d + tT$ preserves h, $S(d) \pmod{r_i}$, and $V_d \pmod{q_i}$, by Lemma 3. Hence the same character argument works for all $t \geq 0$. The rows $d = 12, 15, 27, 243$ give exactly (6). □

Corollary 3 (Exact local power criteria)**.** *Put $Q = 440334654777631$ and let u be a positive integer. For every rotation at the indicated boundary,*

d	*equivalence over the stated field*
$27u$	*R is a fifth power in $\mathbb{Q}_Q$ $\iff$ u is a fifth power in $\mathbb{Q}_Q$*
$243u$	*R is a seventh power in $\mathbb{Q}_{757}$ $\iff$ u is a seventh power in $\mathbb{Q}_{757}$.*

Explicitly, for $(p, e) = (Q, 5)$ or $(757, 7)$, write $u = p^a t$ with $p \nmid t$. The local condition is exactly

$$e \mid a, \qquad t^{(p-1)/e} \equiv 1 \pmod{p}. \tag{17}$$

If $3 \nmid u$, failure of this condition excludes every integer perfect power in the whole rotation block.

Proof. Theorem 4 gives $v_p(R) = v_p(u) = a$ and

$$R/p^a \equiv 10^{D(k)} 178343635184937\, t \pmod{Q}, \qquad R/p^a \equiv 10^{D(k)} 300t \pmod{757},$$

respectively. The displayed coefficients and 10 have character 1 for exponents 5 and 7 respectively, as checked in the certificates. Since $e \neq p$, Hensel's lemma lifts every nonzero eth root modulo p. Thus a number in $\mathbb{Q}_p$ is an eth power exactly when its valuation is divisible by e and its normalized unit is an eth power modulo p. This proves both equivalences. When $3 \nmid u$, the only possible prime exponent is 5 or 7 respectively, by Theorem 2. □

Proof of Corollary 2. The 3-adic restrictions leave only the following disjoint sets:

$$d \equiv 2 \pmod 3, \qquad d \equiv 12, 15 \pmod{27}, \qquad d \equiv 18 \pmod{27}, \qquad 27 \mid d.$$

In the first and third sets, a perfect power must be a square; in the second, a cube. Put $Q = 440334654777631$. In every fixed class modulo 27, Lemma 3 makes V_d affine modulo each of $37, 333667, 757, Q$. In the class $d \equiv a \pmod{27}$ the respective slopes are $1, 331988, 538, 178343635184937$ multiplied by $10^{S(a)}$; all are nonzero. The four residues are independent by the Chinese remainder theorem. For a square candidate the exact valuation of $S(d)$ from (11) shows $27 \mid S(d)$. Thus all four rotation characters are fixed, and the proportion of residues that are zero or squares at all four primes is

$$\alpha = \frac{19}{37}\frac{166834}{333667}\frac{379}{757}\frac{Q+1}{2Q}.$$

For cubes use 37 and 333667, giving the proportion $\beta = (13/37)(111223/333667)$, as in (4). The upper density of the remaining d is therefore at most $(10\alpha + 2\beta + 1)/27$.

Inside $27 \mid d$, Corollary 3 excludes further disjoint sets of densities

$$\delta_5 = \frac{2}{81}\frac{4(Q-1)}{5Q}, \qquad \delta_7 = \frac{2}{729}\frac{6 \cdot 756}{7 \cdot 757}.$$

Indeed $v_3(d)$ is respectively 3 or 5, and nonzero nonpowers occupy 4/5 or 6/7 of the corresponding finite-field units. Also $d = 81u$ with $u \equiv 1, 7 \pmod 9$ is excluded by Corollary 1, adding the disjoint density $2/729$. Consequently

$$\underline{\mathrm{dens}}\{d : \text{every boundary rotation is excluded}\} \geq 1 - \frac{10\alpha + 2\beta + 1}{27} + \delta_5 + \delta_7 + \frac{2}{729} = \rho. \qquad \square$$

5 A uniform sieve for powerful values

We now prove Theorem 1. In a fixed d-digit interval set

$$B = g^d, \quad L = g^{d-1}, \quad c = B - 1, \quad A = c^2 W_{L-1} + Lc + 1.$$

The recurrence $W_{m+1} = BW_m + (m+1)$ gives, for $L - 1 \leq m \leq B - 1$,

$$c^2 W_m = AB^{m-L+1} - ((m+1)c + 1), \qquad D(m) = D(L-1) + d(m - L + 1).$$

For fixed k, writing $u = g^{D(k)}$, substitution into (1) gives

$$R_{m,k} = \alpha B^m + \beta m + \gamma, \quad \begin{cases} \alpha = (uA/c^2 - g^{D(L-1)} W_k) B^{1-L}, \\ \beta = -u/c, \quad \gamma = W_k - uB/c^2. \end{cases} \tag{18}$$

At every prime $p \nmid g(B-1)$ the coefficients are defined modulo p^2 and β is a unit. The right side provides a periodic modular extension; only its values inside the digit interval are identified with the concatenation.

Lemma 4 (Valuation-one densities)**.** *Let $p \nmid B\beta$ and $F(n) = \alpha B^n + \beta n + \gamma \pmod{p^2}$. The event $p \mid F(n)$, $p^2 \nmid F(n)$ has period dividing $T_p = p^2(p-1)$ and density $\rho_p = (p-1)/p^2$. For any finite set P of eligible primes the events have joint density $\prod_{p \in P} \rho_p$ in a period dividing $\mathrm{lcm}_{p \in P} T_p$.*

Proof. Fix $n \pmod{p-1}$. The exponential term is constant modulo p, and the linear coefficient is nonzero, so exactly one class $n \pmod p$ is a root. Its p lifts differ by $p(p-1)t$. Euler's theorem fixes the exponential term modulo p^2 on these lifts, while the linear term takes p distinct values. Precisely one lift is zero modulo p^2, leaving $p - 1$ of valuation one.

For the joint assertion induct on the largest prime q. The smaller-prime events have a period T with no prime divisor as large as q. Thus $q \nmid H = \mathrm{lcm}(T, q-1)$. In each fixed class modulo H, the smaller-prime events are fixed and the Chinese remainder theorem lets $n \pmod{q^2}$ vary freely. The local count gives the same fraction ρ_q in every class. Averaging multiplies the previous joint density by ρ_q. This allows overlapping periods, including a smaller prime dividing a larger prime minus one. $\square$

For an interval I of H integers and a subset P', Lemma 4 gives its simultaneous event count as $H\prod_{p\in P'}\rho_p + O(\prod_{p\in P'}p^3)$, since the interval error is at most the joint period. Take eligible primes at most z, put $\lambda=\sum\rho_p$, and choose an even integer $K\ge 0$. Powerful values avoid every valuation-one event. Finite Bonferroni gives

$$\#\{n\in I : F(n) \text{ powerful}\} \le H\left(e^{-\lambda}+\frac{\lambda^{K+1}}{(K+1)!}\right)+(K+1)z^{4K}. \tag{19}$$

Indeed, if e_j are the elementary symmetric sums of the ρ_p, the even truncated main term is at most $\prod(1-\rho_p)+e_{K+1}\le e^{-\lambda}+\lambda^{K+1}/(K+1)!$. There are at most z^j subsets of size j, each costing at most z^{3j}, which proves the displayed error. These estimates are uniform in the coefficients of (18) and in k.

Choose

$$\eta=\frac{\log g}{100},\qquad z=\exp\left(\frac{\eta d}{\log d}\right),\qquad K=2\lceil 4\log d\rceil,$$

and use primes $d^2<p\le z$ with $p\nmid g(g^d-1)$. For large d, $z>d^2$. Omitted primes exceeding d^2 and dividing g^d-1 number at most $\log(g^d-1)/(2\log d)$, with reciprocal sum $O_g(1/(d\log d))$. The fixed prime divisors of g eventually lie below d^2. Mertens' estimate [3] therefore yields

$$\lambda=\log\log z-\log\log(d^2)+O_g(1)=\log d-2\log\log d+O_g(1).$$

In particular $0<\lambda\le\log d$ eventually, so $e^{-\lambda}\ll_g(\log d)^2/d$. Also

$$\frac{\lambda^{K+1}}{(K+1)!}\le(e/8)^{K+1}\le d^{-8},\qquad (K+1)z^{4K}<g^{d/2}$$

for large d; the latter follows from $K\le 10\log d$ and $\log((K+1)z^{4K})\le\log(K+1)+40\eta d<d\log g/2$. Thus for every fixed k and every interval inside the digit block with $m>k$,

$$\#\{m\in I : R_{m,k} \text{ powerful}\}\ll_g |I|\frac{(\log d)^2}{d}+g^{d/2}. \tag{20}$$

Let D be the digit length of $\lfloor X\rfloor$. For each $d\le D$ and k, apply (20) on $\max(g^{d-1},k+1,2)\le m\le\min(X,g^d-1)$. There are at most g^d choices of k, so the total interval error is $\sum_{d\le D}g^{3d/2}=O_g(X^{3/2})$. Blocks with $d\le D/2$ contain only $O_g(X)$ pairs and may be bounded trivially. On the other blocks the main factor is $O((\log D)^2/D)$, and there are $O(X^2)$ pairs. This proves (2).

Finally, each m supplies m distinct rotations. If m has $d\ge 2$ digits, the cyclic digit word has a unique occurrence of 1 followed by $d-1$ zeros: it is the integer g^{d-1}. Such a zero run cannot cross an integer boundary or the cyclic join, whose next digit is 1. The unique marker prevents a nonidentity digit rotation from fixing the word. For $d=1$ all digits are distinct. Digit lengths increase strictly with m, so the sequence through block $m=X$ has $X(X+1)/2$ entries. Bounding a partial final block by the whole block and taking $X\asymp\sqrt{N}$ proves the sequence-index assertion. □

References

[1] OEIS Foundation, Sequence A001292, https://oeis.org/A001292, accessed October 27, 2026.

[2] M. Ripà, *On some open problems concerning perfect powers*, arXiv:2205.10163, version 4 (2024), https://arxiv.org/abs/2205.10163.

[3] F. Mertens, Ein Beitrag zur analytischen Zahlentheorie, *J. reine angew. Math.* **78** (1874), 46–62, https://doi.org/10.1515/crll.1874.78.46.